

and $\mathbb{P}((\sqrt{\Gamma_X}B_1, \sqrt{\Gamma_Y}B_2) \in \text{Dis}(F_2)) = 0$, we can therefore apply the continuous mapping theorem to obtain

$$F_2(H_m, J_n) \Rightarrow F_2(\sqrt{\Gamma_X}B_1, \sqrt{\Gamma_Y}B_2). \quad (3.11)$$

Next, we show that $F_2(H_m, J_n)$ approximates the asymptotic law of the integrand (3.8) in the sense that

$$d_1((\tilde{Z}_{m,n} - (\cdot)^2 Z_{m,n})^2, F_2(H_m, J_n)) \rightarrow 0 \quad (3.12)$$

in probability, where d_1 denotes the Skorokhod metric on D_1 , cf. (3.30) for an exact definition. The Skorokhod metric d_1 satisfies

$$d_1(u, v) < \delta \text{ implies } d_1(u^2, v^2) < (1 + \|u\|_\infty + \|v\|_\infty)\delta$$

for any $u, v \in D_1$ and $\delta > 0$. So, in order to verify (3.12), it is sufficient that

$$d_1(\tilde{Z}_{m,n} - (\cdot)^2 Z_{m,n}, G_1(H_m, J_n) - (\cdot)^2 F_1(H_m, J_n)) \rightarrow 0 \text{ in probability.} \quad (3.13)$$

To that end, we first use the triangle inequality to bound the left-hand side in (3.13) by

$$\begin{aligned} & d_1(\tilde{Z}_{m,n} - (\cdot)^2 Z_{m,n}, G_1(H_m, J_n) - (\cdot)^2 Z_{m,n}) \\ & + d_1(G_1(H_m, J_n) - (\cdot)^2 Z_{m,n}, G_1(H_m, J_n) - (\cdot)^2 F_1(H_m, J_n)). \end{aligned} \quad (3.14)$$

We begin with the first term in (3.14). We have by the definition of the Skorokhod metric

$$\begin{aligned} & d_1(\tilde{Z}_{m,n} - (\cdot)^2 Z_{m,n}, G_1(H_m, J_n) - (\cdot)^2 Z_{m,n}) \\ & = \inf_{\lambda \in \Lambda} \left\{ \max \left\{ \sup_s |\tilde{Z}_{m,n}(s) - s^2 Z_{m,n} - G_1(H_m, J_n)(\lambda(s)) + \lambda(s)^2 Z_{m,n}|, \right. \right. \\ & \quad \left. \left. \sup_s |\lambda(s) - s| \right\} \right\} \\ & \leq \inf_{\lambda \in \Lambda} \left\{ 2(2|Z_{m,n}| + 1) \max \left\{ \sup_s |\tilde{Z}_{m,n}(s) - G_1(H_m, J_n)(\lambda(s))|, \sup_s |\lambda(s) - s| \right\} \right\} \\ & = 2(2|Z_{m,n}| + 1) d_1(\tilde{Z}_{m,n}, G_1(H_m, J_n)). \end{aligned}$$

In the second step we have already established $Z_{m,n} \Rightarrow \xi B(1)$. Furthermore, it follows that $d_1(\tilde{Z}_{m,n}, G_1(H_m, J_n)) \rightarrow 0$ in probability. Indeed, we have

$$\begin{aligned} d_1(\tilde{Z}_{m,n}, G_1(H_m, J_n)) & = d_1(\tilde{Z}_{1,m,n}, \tilde{Z}_{2,m,n}, G_{1,1}(H_m, J_n), g_{1,2}) \\ & \leq \sup_{s \in [0,1]} \left| \tilde{Z}_{1,m,n}(s) - G_{1,1}(H_m, J_n)(s) \right| \sup_{s \in [0,1]} |\tilde{Z}_{2,m,n}(s)| \\ & \quad + \sup_{s \in [0,1]} |G_{1,1}(H_m, J_n)(s)| \sup_{s \in [0,1]} \left| \tilde{Z}_{2,m,n}(s) - g_{1,2}(s) \right|. \end{aligned}$$

On the one hand,

$$\left(\sup_{s \in [0,1]} |\tilde{Z}_{2,m,n}(s)| \right)_{m,n} \quad \text{and} \quad \left(\sup_{s \in [0,1]} |G_{1,1}(H_m, J_n)(s)| \right)_{m,n}$$

are tight real-valued sequences. On the other hand, it is true that in probability

$$\sup_{s \in [0,1]} |\tilde{Z}_{1,m,n}(s) - G_{1,1}(H_m, J_n)(s)| \rightarrow 0, \quad (3.15)$$

$$\sup_{s \in [0,1]} |\tilde{Z}_{2,m,n}(s) - g_{1,2}(s)| \rightarrow 0. \quad (3.16)$$

Indeed, (3.15) is bounded above as follows

$$\begin{aligned} & \sup_{s \in [0,1]} |\tilde{Z}_{1,m,n}(s) - G_{1,1}(H_m, J_n)(s)| \\ & \lesssim \sup_{s \in [0,1]} s^{1-\gamma} \left| \frac{sm/\lfloor sm \rfloor}{\sqrt{m/(m+n)}} - \frac{1}{\sqrt{\tau}} \right| |H_m(s)| \\ & \quad + \sup_{s \in [0,1]} s^{1-\gamma} \left| \frac{sn/\lfloor sn \rfloor}{\sqrt{n/(m+n)}} - \frac{1}{\sqrt{1-\tau}} \right| |J_n(s)| \rightarrow 0 \end{aligned}$$

in probability. And similarly for (3.16)

$$\sup_{s \in [0,1]} |\tilde{Z}_{2,m,n}(s) - g_{1,2}(s)| \lesssim \sup_{s \in [0,1]} s^\gamma \frac{\sqrt{m}}{\lfloor ms \rfloor} |H_m(s)| + \sup_{s \in [0,1]} s^\gamma \frac{\sqrt{n}}{\lfloor ns \rfloor} |J_n(s)| \rightarrow 0$$

in probability by Lemma 3.2 – using the assumed strong law of large numbers for $m^{-1}\bar{X}_m(1)$ and $n^{-1}\bar{Y}_n(1)$. Hence by Slutsky's theorem the first term in (3.14) converges to zero in probability.

The second term in (3.14) can be bounded above by choosing the identity in the definition of the Skorokhod metric, i.e.,

$$\begin{aligned} & d_1(G_1(H_m, J_n) - (\cdot)^2 Z_{m,n}, G_1(H_m, J_n) - (\cdot)^2 F_1(H_m, J_n)) \\ & \leq \sup_s |s^2 Z_{m,n} - s^2 F_1(H_m, J_n)| = |Z_{m,n} - F_1(H_m, J_n)|. \end{aligned}$$

Then once more, relying on the results of step two, the right-hand side tends to zero in probability. Thus, (3.12) is established and together with (3.11) we conclude by Slutsky's theorem

$$\tilde{Z}_{m,n} - (\cdot)^2 Z_{m,n} \Rightarrow F_2(\sqrt{\Gamma_X} B_1, \sqrt{\Gamma_Y} B_2).$$

Also, it is straightforward to see

$$\begin{aligned} & F_2(\sqrt{\Gamma_X} B_1, \sqrt{\Gamma_Y} B_2)(s) \\ & \stackrel{\mathcal{L}}{=} \left(2s(\mathbb{E}[X_0] - \mathbb{E}[Y_0]) \sqrt{\frac{\Gamma_X}{\tau} + \frac{\Gamma_Y}{1-\tau}} (B(s) - sB(1)) \right)^2 = \xi^2 s^2 (B(s) - sB(1))^2. \end{aligned}$$

Finally, since integration is a continuous functional on D_1 (w.r.t. d_1), we can deduce on the one hand

$$\left\{ \int_0^1 F_2(H_m, J_n)(s)^2 \nu(ds) \right\}^{1/2} \Rightarrow \xi \left\{ \int_0^1 s^2 (B(s) - sB(1))^2 \nu(ds) \right\}^{1/2}.$$

And on the other hand,

$$\sqrt{m+n} \tilde{V}_{m,n} - \left\{ \int_0^1 F_2(H_m, J_n)(s)^2 \nu(ds) \right\}^{1/2} \rightarrow 0 \quad (3.17)$$

in probability.

Step 4. It is just a formality to remark that the weak convergence which we established in steps two and three carries over to the two dimensional setting: Indeed,

$$\begin{aligned} & \left(F_1(H_m, J_n), \left\{ \int_0^1 F_2(H_m, J_n)(s)^2 \nu(ds) \right\}^{1/2} \right) \\ & \Rightarrow (F_1(\sqrt{\Gamma_X} B_1, \sqrt{\Gamma_Y} B_2), \left\{ \int_0^1 F_2(\sqrt{\Gamma_X} B_1, \sqrt{\Gamma_Y} B_2)(s)^2 \nu(ds) \right\}^{1/2}) \\ & \stackrel{\mathcal{L}}{=} \left(\xi B(1), \xi \left\{ \int_0^1 s^2 (B(s) - sB(1))^2 \nu(ds) \right\}^{1/2} \right) \end{aligned}$$

and the convergence results in (3.10) and (3.17) remain true as well when considered jointly in a vector. This completes the proof. $\square$

Lemma 3.2. *Let $(Z_n)_n$ be a sequence of identically distributed, integrable random variables and assume that the corresponding partial sum $\overline{Z}_k = \sum_{i=1}^k Z_i$ satisfies a SLLN, i.e., $k^{-1} \overline{Z}_k \rightarrow \mathbb{E}[Z_0]$ almost surely. Then for all $\delta, \varepsilon > 0$ and $\gamma \in (0, 1)$ there is a constant $M \in \mathbb{N}$ such that*

$$\mathbb{P} \left(\max_{k \leq m} \left(\frac{k+1}{m} \right)^\gamma |k^{-1} \overline{Z}_k - \mathbb{E}[Z_0]| > \varepsilon \right) < \delta \quad \forall m \geq M. \quad (3.18)$$

Proof of Lemma 3.2. By assumption there is a measurable set $\Omega_0 \subseteq \Omega$ with $\mathbb{P}(\Omega_0) = 1$ and

$$|k^{-1} \overline{Z}_k(\omega) - \mathbb{E}[Z_0]| \rightarrow 0 \quad \forall \omega \in \Omega_0.$$

Hence, for each $\omega \in \Omega_0$, there exists a $M_0(\omega) \in \mathbb{N}$ such that

$$|k^{-1} \overline{Z}_k(\omega) - \mathbb{E}[Z_0]| < \frac{\varepsilon}{2^\gamma} \quad \forall k \geq M_0(\omega). \quad (3.19)$$

For $\omega \in \Omega \setminus \Omega_0$ set $M_0(\omega) = 0$. Then $M_0: \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ is almost surely bounded and therefore we can find a constant $M_1 \in \mathbb{N}$ such that $\mathbb{P}(M_0 > M_1) \leq \delta/2$.